Input Assumptions

shg_mode_survey · loss model + material dispersion, referenced by every results report

Every per-sweep PDF this project produces should cite this document rather than re-stating or re-deriving any assumption. Two bands matter throughout: SH = 215–235 nm (deep UV), pump = 430–470 nm (true blue) — shaded orange and blue respectively in every chart below.

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STATUS AT TIME OF WRITING

AlN, SiNx, TiO2 dispersion: committed, wired into at least one geometry family.
AlN's own UV k(lambda): a real MOCVD-measured candidate was picked (AM187-LT6, section 3) -- k_e >> k_o, ~0.04-0.05 at SH, ~0.012 at pump. Wired as an opt-in mechanism (geom.params ['aln_lossy']) and used to REPROCESS completed runs, but NOT adopted as the working assumption at either band: its SH-band implication (~100,000 dB/cm) was checked against real device output and found 3-4 orders of magnitude too high (section 3b). SH loss instead uses a flat, empirically-calibrated 100 dB/cm placeholder; pump loss stays at its prior native (scattering + interface-defect mechanism) value, unchanged.
ScAlN: index fitted (8 compositions, Baumler 2019). UV k and chi(2) d33/d31 vs. Sc fraction are disputed between sputtered and MBE literature — see section 6.
GaAlN: no dispersion fit exists yet. Section 5 documents the gap, not a fix.

0. Figures of merit: what NCE and peak efficiency mean

A phase-matched crossing

The pump mode's effective index n_eff,pump(2*lambda) equals the SH mode's n_eff,SH(lambda) -- pump and SH stay in step as they co-propagate, so generated SH keeps adding constructively instead of periodically cancelling. Every crossings.csv row is one such pump/SH mode pair.

NCE -- eta_pct_per_W_cm2 (normalized conversion efficiency)

A pure field-overlap quantity: how well the pump-driven nonlinear polarization (chi(2):E_pump^2) lines up with the SH mode's own field shape, for 1W-normalized modes (%/W/cm^2 -- shg_physics.eta_norm_per_W_m2/eta_to_pct_per_W_cm2). Computed from the two mode PROFILES at one cross-section -- propagation distance and device length never enter. overlap_shape (0-1) is the companion dimensionless version.

Loss-limited peak efficiency -- peak_efficiency_pct_per_W

What you'd actually measure from a real device of length L, pump/SH power-loss coefficients alpha_p/alpha_s (shg_physics.loss_limited_length):

```
P_SH(L) = eta * P_pump^2 * L_eff(L)^2
L_eff(L) = [exp(-alpha_p*L) - exp(-alpha_s*L/2)] / (alpha_s/2 - alpha_p)
```

L_eff -> L as loss -> 0 (plain quadratic P_SH ~ L^2); with real loss it rises, peaks at an optimum length L_opt (L_opt_mm), then DECLINES for longer devices -- not a plateau. peak_efficiency_pct_per_W = eta * L_eff(L_opt)^2: 'is this design any good' for a real device, as opposed to eta alone.

Why NCE never moves when the loss model changes

eta and L_eff come from disjoint inputs -- eta from the transverse mode SHAPES alone (no z-dependence), L_eff from alpha_p/alpha_s alone (never looks at mode shape). Changing the loss model -- AlN's own absorption, sidewall roughness, a different cladding -- can only move L_eff (hence L_opt and peak efficiency); it has no path back to eta. Every loss-comparison report here (e.g. aln_loss_report.pdf) shows NCE exactly unchanged between scenarios, while peak efficiency moves -- by construction.

1. Loss model: scattering + absorption

Scattering (native EMode, Volume Current Method)

Sidewall (vertical): 2.5 nm RMS roughness, 50 nm correlation length.
Top/bottom (horizontal): 0.3 nm RMS roughness, 10 nm correlation length.
Values as currently used in survey_config.py. The "Loss Model Summary" memo (see section 2) used 2 nm sidewall RMS instead — everything else matches.

Absorption (non-uniform, growth-interface defects)

Each mechanism: a local power-absorption coefficient (alpha0, dB/m) at a physical boundary, decaying into the core (exponential or step) over a penetration depth L (nm). Modal loss is the Poynting-flux-weighted average of that profile over the mode. Currently active in survey_config.py's pump_absorption_mechanisms (4 of them; sh_absorption_mechanisms is empty — this interface-defect model contributes ZERO absorption at the SH wavelength). SH loss's own flat, confinement-weighted 100 dB/cm calibrated absorption is a SEPARATE mechanism, not part of this interface-defect list -- see section 3b:

Dislocations `2000 dB/m, exp, 40nm`
Chen et al., Opt. Express 27(12), 17262 (2019) — but
literature treats this as SCATTERING with a TM>TE dependence, not modeled here as such.

C impurities `3500 dB/m, step, 50nm`
Robinson et al., arXiv:2606.19615 (2026)

Si impurities `3500 dB/m, step, 50nm`
No literature value found.
Engineering placeholder.

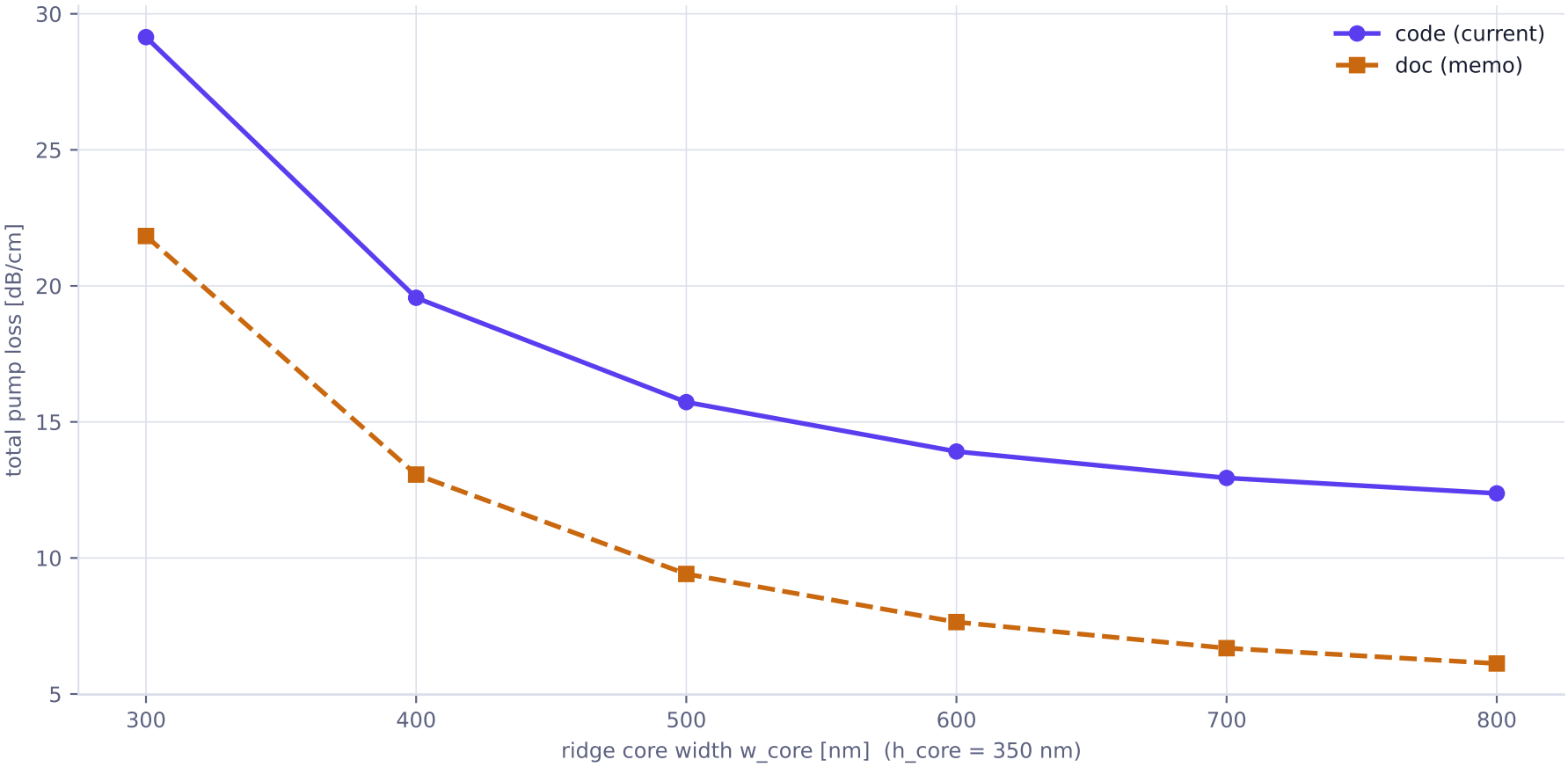
O impurities `5000 dB/m, step, 50nm`
Bickermann et al., Appl. Phys. Lett. 95, 262104 (2009)

Not currently modeled at all (zero sidewall mechanisms in survey_config.py):

Oxidation ingress, sidewall contamination, sidewall damage — all three have no literature support either; the Loss Model Summary memo carries two of them at nonzero values (see section 2's comparison). Dropped here, not evaluated as zero-impact — just never added.

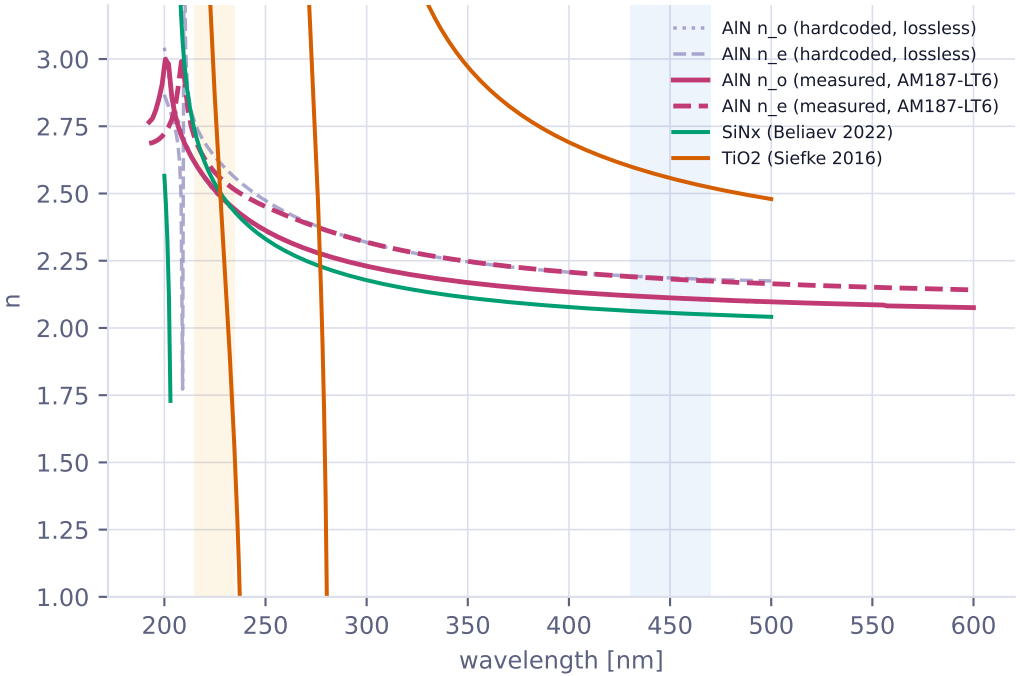
2. Pump-loss assumption comparison

code = survey_config.py's current 4-mechanism set (section 1). doc = the "Loss Model Summary & Results" memo's own "Actual Inputs Used" table (7 mechanisms, different roughness). Computed from trial_grid's already-exported TM00 pump fields (ridge, h_core=350 nm fixed, no new EMode calls). Scattering's roughness difference (2.5 vs 2.0 nm sidewall) is applied as an analytic (RMS)^2 scaling to the combined EMode-native value, not a fresh VCM solve — a rougher approximation than the absorption-side comparison; see the script's docstring.

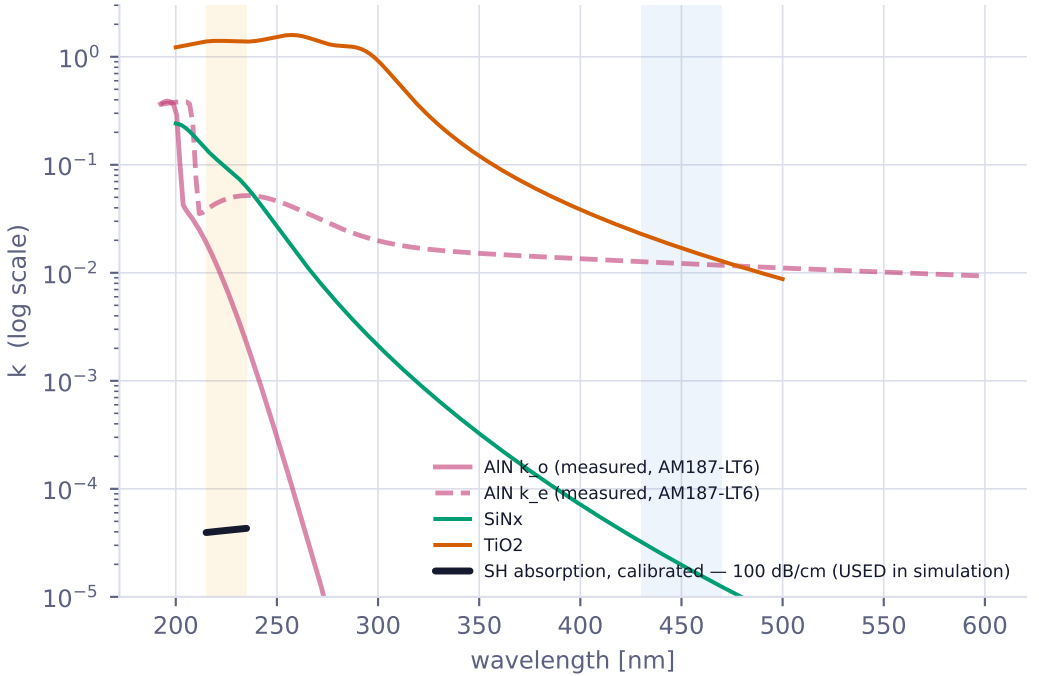


doc/code ratio across this width range: 0.49–0.75x — doc consistently LOWER than code, not higher, despite its extra sidewall terms (its exponential-decay, shorter-penetration-depth choices for C/Si/O more than offset that). Not an order-of-magnitude gap. Per 2026-10-12 direction: logged here, current code assumptions kept as-is.

3. Core/cladding material dispersion



AlN's measured k (AM187-LT6) is NOT used for SH loss — it implied ~100,000 dB/cm, 3-4 orders of magnitude above what fabricated devices actually permit (section 3b)



AlN measured curves: sample AM187 (MOCVD, new model, 6.4.26), position LT(6) -- the lowest, non-negative k_e among all MOCVD-labeled positions in the team's own ellipsometry repo (github.com/uviquityBF/ellipsometry_data), selected 2026-10-12. Shown for reference only (right panel, log scale) -- NOT what the simulation actually uses for SH loss; see section 3b.

3b. What SH loss the simulation actually uses

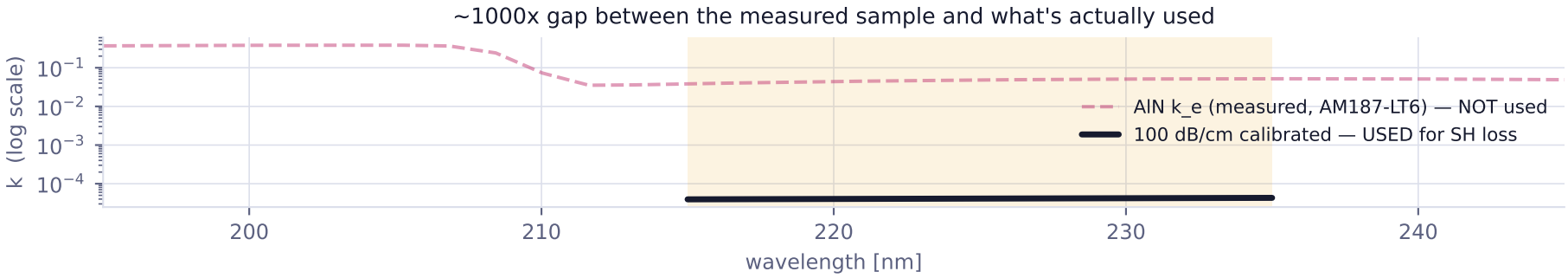
AlN's own measured k_e (AM187-LT6, section 3) implies roughly 100,000 dB/cm of SH-band absorption for a representative crossing (ridge h=350 w=500nm, TM00->TM04 -- the closest match in this project's dataset to real fabricated/tested devices). That number is NOT used -- it was checked against real device measurements and found to be 3-4 orders of magnitude too high.

How the empirical bound was derived (2026-10-12)

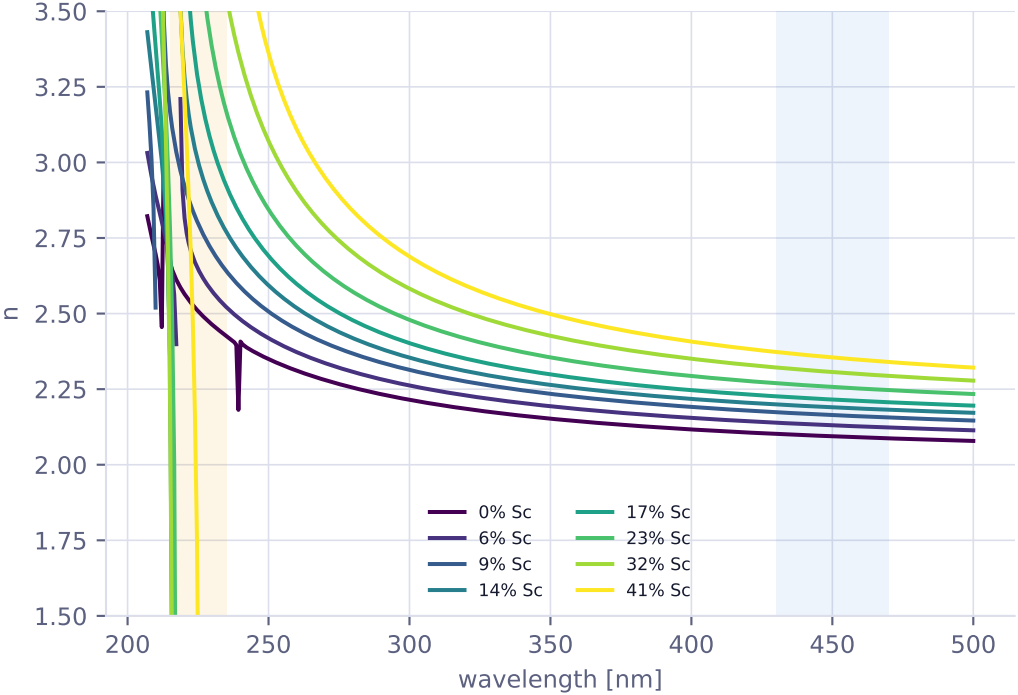
A fabricated device (same geometry) measures ~200 nW SH output at best, ~10s of nW typically, under ~200 mW incident pump power (10-50% coupling efficiency into the waveguide, unmeasured) over a ~2 mm interaction length. Inverting this project's own loss-limited-efficiency model (shg_physics.loss_limited_length, evaluated at the real fixed 2mm length -- not an idealized optimum) for the SH loss consistent with that output bounds alpha_SH to roughly 15-680 dB/cm across the coupling-efficiency uncertainty -- nowhere near the ~100,000 dB/cm the ellipsometry sample implies.

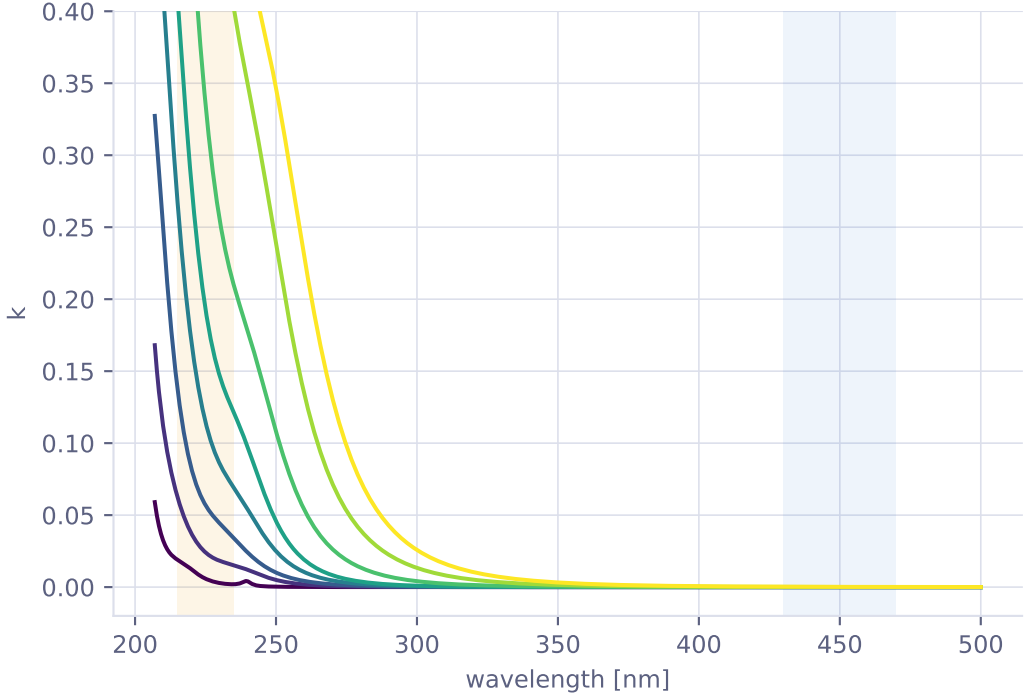
Working value: 100 dB/cm, confinement-weighted (2026-10-12 decision)

100 dB/cm is a round number inside that empirically-bounded range -- not a measurement of any specific physical mechanism (absorption, under-modeled scattering, or something else is equally plausible as the real cause). Applied as geometry.sh_absorption_calibrated_dB_per_m, confinement-weighted by the mode's power fraction in the AlN core exactly like AlN's own absorption would be (geom.core_mask) -- see survey.py's bulk_absorption_loss. Flat across the SH band (215-235nm) by construction, since it's a single working value rather than a fitted spectral shape. Applies ONLY to SH loss, never to pump -- it was back-calculated purely from SH output. EMode-native sidewall-roughness scattering at SH (already in the model throughout, unaffected by any of this -- see section 1) is ADDED on top, not replaced.



4. ScAlN composition sweep (index ready, k/d33 open)





Source: Baumler et al., J. Appl. Phys. 126, 045715 (2019), Fig. n/k; digitized by digitize_baumler.py (k resolution ~0
k resolution ~0.008, fit to 0 beyond 300.0 nm (digitization limit, not a physical statement) — UV k and the chi(2) d33/d31-vs-Sc
relationship are NOT from this fit and remain disputed; see section 6.

5. GaAlN: not yet sourced

No n(lambda)/k(lambda) dispersion fit exists for GaAlN (Al_xGa_1-xN) in this repo. This page documents the gap and what's known from a first literature pass, not a resolution.

Two independent motivations were identified for keeping GaAlN in the design matrix even with d33 enhancement deprioritized (user direction, 2026-10-12):

1. Index engineering for a Cerenkov cladding (composition tunes index, not nonlinearity).
2. As a loading strip that reshapes the AlN mode — pulling it from the substrate interface, or improving the AlN region's own overlap — without needing the strip itself active.

Composition-dependent d33 (NIST, Sanford et al.): decreases with increasing Al fraction x, with a SIGN CHANGE near x ~ 0.65. The favorable (higher-d33) direction is toward Ga-rich compositions — but GaN's bandgap (~3.4 eV) is well below the SH photon energy at 215-235 nm (~5.5 eV), so Ga-rich AlGaN is almost certainly strongly absorbing exactly where this project needs transparency. The useful composition window (enough Ga for index tuning, not so much that it absorbs the SH) is likely narrow and has not been checked quantitatively.

Needed before GaAlN can enter any sweep: n(lambda)/k(lambda) across 215-470 nm for a composition range still TBD (narrow, Al-rich, pending the transparency check above).

6. Nonlinear (chi(2)) tensor summary

wurtzite 6mm, c || y. Nonzero: d33; d31 = d32; d15 = d24. Py = eps0[d31(Ex^2+Ez^2) + d33 Ey^2], Px = 2 eps0 d15 Ex Ey, Pz = 2 eps0 d15 Ez Ey. Kleinman symmetry (d15=d31) assumed unless data says otherwise -- questionable near the SH band edge, treat d15 as separate.

AlN, bulk Majkic 2017, pssb 201700077	d33=4.3 +/- 0.3 (@1030nm) pm/V	d31=~0.1
AlN (working value) user's current value -- used throughout this project	d33=4.7 pm/V	d31=~0.1
AlN, sputter film Yoshioka 2021, APL Mater. 9, 101104	d33=5.1 +/- 0.4 pm/V	d31=0.07
ScAlN sputter, 10/20/28/36% Sc Yoshioka 2021	d33=15.8/42.5/46.9/62.3 pm/V	d31=0.84/2.4/3.4/4.5
ScAlN MBE on sapphire, 10/15/20/25% Lee et al., arXiv 2607.14590 (2026)	d33=-2.5/5.1/-3.1/3.2 (noisy) pm/V	d31=1.6/3.3/4.9/1.6

DISAGREEMENT

Sputtered films show strong, clean d33 enhancement with Sc content. MBE films (direct on sapphire, bypassing the AlN buffer) show d33 scattered/suppressed instead, with d31 showing the strong, clean enhancement there. Neither growth method has been measured anywhere near 215-235nm/430-470nm (both cited studies: ~1030nm pump). Implication: if d33 doesn't actually enhance for your growth process, a ScAlN core's real lever may be d31 (TE-pump -> TM-SH coupling, Py = d31*Ex^2), not d33 -- a different mode/polarization design than naively assuming 'bigger d33'. overlap_terms() in shg_physics.py already separates d33/d31/d15 contributions per crossing, so this is mechanically supported once real tensor values are chosen -- the open question is which values, not whether the code can use them.

7. Notes & provenance

Loss model source: "Loss Model Summary & Results", internal memo, Google Drive, created 2026-09-22 (shorthand used throughout this project: [Loss Model Summary, 2026-09-22]). Companion doc: "AlN_Absorption_Literature_Review" (same Drive folder).

Material fits: materials/fit_sinx_sellmeier.py (Beliaev et al., Thin Solid Films 763, 139568, 2022), materials/fit_tio2_sellmeier.py (Siefke et al., Adv. Opt. Mater. 4, 1780, 2016), materials/fit_baumler_sellmeier.py (Baumler et al., J. Appl. Phys. 126, 045715, 2019, digitized). AlN o/e equations hardcoded in geometry.py (AlN_EQ_O/AlN_EQ_E) -- real-valued, lossless; deliberately kept unchanged for continuity with prior results (2026-10-12 decision) rather than switched to the measured n below -- see section 3b for where k (a separate, independent addition) is actually used instead.

AlN measured n/k: materials/AlN_AM187_LT6_measured_nk.csv, exported 2026-10-12 from github.com/uviquityBF/ellipsometry_data (a sibling repo, cloned to ..\ellipsometry_data, not inside this one) -- sample AM187 (MOCVD, new model, 6.4.26), wafer position LT(6), picked as the lowest non-negative k_e among all MOCVD-labeled positions in that repo's catalog (240 anisotropic positions checked). Raw WVASE ellipsometry export; k_e fit separately to a 3-Lorentzian form (materials/fit_aln_measured_k.py, decoupled from n) for simulation use -- geometry.AlN_K_E_EQ / aln_core_loss_dB_per_m.

SH-loss calibration (section 3b): geometry.SH_ABSORPTION_CALIBRATED_DB_PER_CM = 100, chosen 2026-10-12 against real fabricated-device measurements (user-supplied, not a literature or ellipsometry source) -- ~200nW best-case / ~10s of nW typical SH output, ~200mW incident pump power (10-50% coupling efficiency, unmeasured), ~2mm interaction length, ridge h=350 w=500nm geometry (closest dataset match). Derivation: inverting shg_physics.loss_limited_length at the real fixed device length for the SH loss consistent with that output. See PLAN.md's 2026-10-12 "empirical SH-loss bound" entries for the full numeric derivation (per-coupling-scenario table) this report's section 3b summarizes.

Pump-loss comparison: compare_pump_loss_assumptions.py, computed from trial_grid's already-exported TM00 pump fields -- no new EMode calls. Re-run that script directly to regenerate runs/pump_loss_assumption_comparison.csv.